

The three Kanade–Russell identities modulo nine

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Abstract

We prove the three symmetric Kanade–Russell identities modulo nine.

1 Introduction

Write

$$(a; q)_n = \prod_{j=0}^{n-1} (1 - aq^j), \quad (a_1, \dots, a_s; q)_\infty = \prod_{i=1}^s \prod_{j \geq 0} (1 - a_i q^j).$$

Define

$$F(x, y) = \sum_{m, n \geq 0} \frac{q^{m^2 + 3mn + 3n^2} x^m y^n}{(q; q)_m (q^3; q^3)_n}, \quad (A, B, C) = (F(1, 1), F(q, q^3), F(q^2, q^3)). \quad (1.1)$$

Define also

$$\begin{aligned} \text{KR}_1 &= \frac{1}{(q, q^3, q^6, q^8; q^9)_\infty}, \\ \text{KR}_2 &= \frac{1}{(q^2, q^3, q^6, q^7; q^9)_\infty}, \\ \text{KR}_3 &= \frac{1}{(q^3, q^4, q^5, q^6; q^9)_\infty}. \end{aligned} \quad (1.2)$$

Theorem 1.1. *As formal power series in q , and as holomorphic identities on $|q| < 1$,*

$$A = \text{KR}_1, \quad B = \text{KR}_2, \quad C = \text{KR}_3. \quad (1.3)$$

Kanade and Russell used symbolic computation to discover new Rogers–Ramanujan type partition identities and proposed conjectures modulo nine and twelve in 2014 [5]. Subsequent work established several of their conjectures; in particular, Bringmann, Jennings-Shaffer, and Mahlburg proved seven identities modulo twelve [2]. The three partition conjectures modulo nine corresponding to (1.3), however, remained unproved for more than a decade, despite the close resemblance of their partition conditions and product forms to those of the classical Rogers–Ramanujan identities. For work on these conjectures, see, for example, [3, 4, 6, 8, 9, 11, 13–15]. In this paper, we prove (1.3).

Kanade and Russell originally formulated these conjectures as equalities between numbers of partitions [5]. For $i = 1, 2, 3$, let $\mathcal{D}_i$ consist of partitions $\lambda_1 \leq \dots \leq \lambda_\ell$ whose parts are at least i , such that $\lambda_{j+2} - \lambda_j \geq 3$, and such that $\lambda_j + \lambda_{j+1} \equiv 0 \pmod{3}$ whenever $\lambda_{j+1} - \lambda_j \leq 1$. Kanade and Russell conjectured that the number of partitions of each integer in $\mathcal{D}_i$ equals the number with parts congruent to $2^{i-1}, 3, 6, 9 - 2^{i-1} \pmod{9}$. Kurşungöz later expressed the generating functions for $\mathcal{D}_1, \mathcal{D}_2, \mathcal{D}_3$ as the double sums A, B, C , respectively [9, Theorems 7–9]. Thus these partition conjectures are equivalent to (1.3).

Together with the result of Wang and Zhang [15, Theorem 1.1], Theorem 1.1 also implies that the suitably normalized triple (A, B, C) is a weight-zero vector-valued modular function on $\Gamma_0(3)$.

The proof shows that the triple of sums (A, B, C) and the triple of products (KR_1, KR_2, KR_3) give the same value when substituted into the cubic polynomial

$$\mathcal{N}(X, Y, Z) = X^3 + qY^3 - q^2Z^3 + 3qXYZ.$$

Together with the coefficientwise lower bounds $A \geq KR_1$, $B \geq KR_2$, and $C \geq KR_3$, this equality forces the two triples to coincide. Sections 2–4 establish the identities needed to evaluate this polynomial at (A, B, C) . Section 5 evaluates the cubic at the products using the Jacobi addition formula and classical identities for cubic theta and Lambert series. Section 6 combines the resulting equal cubic values with Tsuchioka’s proved coefficientwise inequalities. The first coefficient at which the triples could differ gives a contradiction.

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2 Bilinear identities

Retain the definition of F in (1.1), and define

$$G(x, y) = \sum_{m, n \geq 0} \frac{q^{m^2 - 3mn + 3n^2} x^m y^n}{(q; q)_m (q^3; q^3)_n}.$$

Proposition 2.1 (Bilinear identities). *The following identities hold.*

$$G(x, 1) = F(x, 1)F(x, x^3) + qx^2F(qx, q^3)F(q^2x, q^3x^3), \quad (2.1)$$

$$G(qx, 1) = F(x, 1)F(qx, q^3x^3) - qx F(q^2x, q^3)F(q^2x, q^3x^3), \quad (2.2)$$

$$G(q^{-1}x, q^3) = xF(qx, q^3)F(qx, q^3x^3) + F(q^2x, q^3)F(x, x^3). \quad (2.3)$$

Proof. For each integer j , define

$$R_j = \sum_{n \geq 0} \frac{q^{3n^2 + 3jn}}{(q^3; q^3)_n}.$$

Termwise subtraction and a shift of n give

$$R_j - R_{j+1} = q^{3j+3}R_{j+2}. \quad (2.4)$$

If $g(x) = G(x, 1) = \sum_{j \geq 0} g_j x^j$, then $g_j = q^{j^2} R_{-j} / (q; q)_j$. Equation (2.4) implies, for $j \geq 2$,

$$(1 - q^j)(1 - q^{j-1})g_j = q^{2j-1}(1 - q^{j-1})g_{j-1} + q^{j-1}g_{j-2}.$$

Equivalently,

$$qg(x) - (1 + q + q^2x^2)g(qx) + (1 - q^2x)g(q^2x) + q^2xg(q^3x) = 0. \quad (2.5)$$

For $j \geq 2$, the factor multiplying the coefficient of x^j in this equation is $q(1-q^j)(1-q^{j-1}) \neq 0$. Thus a solution is determined by its constant and linear coefficients. For g , these are

$$g(0) = R_0, \quad [x]g(x) = \frac{q(R_0 + R_1)}{1-q}, \quad (2.6)$$

where $R_{-1} = R_0 + R_1$ follows from (2.4).

The series F satisfies the elementary contiguities

$$\begin{aligned} F(x, y) - F(qx, y) &= qx F(q^2x, q^3y), \\ F(x, y) - F(x, q^3y) &= q^3y F(q^3x, q^6y). \end{aligned} \quad (2.7)$$

Define

$$a(x) = F(x, 1), \quad b(x) = F(qx, q^3), \quad \mathbf{f}(x) = (a(x), b(x), b(qx))^T.$$

Using (2.7) gives

$$\mathbf{f}(qx) = L(x)\mathbf{f}(x), \quad L(x) = \begin{pmatrix} 1 & 0 & -qx \\ 0 & 0 & 1 \\ -x & x & 1 + qx^2 \end{pmatrix}. \quad (2.8)$$

For example, its last row is $b(q^2x) = xb(x) + b(qx) - xa(qx)$.

Define $p(x) = F(qx, x^3)$. It satisfies

$$p(x) = (1 - qx)p(qx) + qx(1 + q + q^2x^2)p(q^2x) + q^4x^2p(q^3x). \quad (2.9)$$

Here is a direct verification. Write $p(x) = \sum c_j x^j$, with $c_j = 0$ for $j < 0$. The coefficient identity is

$$(1 - q^j)c_j = (q^{2j-1}(1 + q) - q^j)c_{j-1} + q^{3j-2}c_{j-2} + q^{2j-3}c_{j-3}.$$

For each summand with $j = m + 3n$, write $X = q^m$, $Y = q^{3n}$. Reindexing the three coefficients on the right by $m - 1, m - 2, n - 1$, respectively, reduces the required equality to

$$1 - XY + \frac{1 - X}{X} - \frac{(1 + q)(1 - X)Y}{q} - \frac{(1 - X/q)(1 - X)Y}{X} - \frac{1 - Y}{X} = 0.$$

This rational identity is exact; the factors $1 - X$, $1 - X/q$, and $1 - Y$ also supply the zero boundary terms.

The column vector $\mathbf{p}(x) = (p(x), p(qx), p(q^2x))^T$ therefore satisfies

$$\mathbf{p}(x) = T(x)\mathbf{p}(qx), \quad T(x) = \begin{pmatrix} 1 - qx & qx(1 + q + q^2x^2) & q^4x^2 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}. \quad (2.10)$$

The contiguities also give

$$\begin{aligned} F(x, x^3) &= p(x) + qxp(qx), \\ F(qx, q^3x^3) &= p(qx) + q^2xp(q^2x), \\ F(q^2x, q^3x^3) &= p(qx). \end{aligned} \quad (2.11)$$

Let $h(x)$ be the right side of (2.1). Then $h(x) = \mathbf{f}(x)^T K_0(x) \mathbf{p}(x)$, where

$$K_0 = \begin{pmatrix} 1 & qx & 0 \\ 0 & qx^2 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Define two further polynomial matrices by

$$K_1 = \begin{pmatrix} 0 & 1 & q^2x \\ 0 & 0 & 0 \\ 0 & -qx & 0 \end{pmatrix}, \quad K_2 = \begin{pmatrix} 1 & qx - 1 & -q^2x \\ 0 & 0 & -q^3x^3 \\ -qx & qx - q^2x^2 & 0 \end{pmatrix}.$$

Direct multiplication gives the three polynomial identities

$$\begin{aligned} K_1(x)T(x) &= L(x)^\top K_0(qx), \\ K_2(x)T(x) &= qxL(x)^\top K_1(qx), \\ (q^2xK_0 - (qx)(1 + q + q^2x^2)K_1 + (1 - q^2x)K_2)T(x) + qxL(x)^\top K_2(qx) &= 0. \end{aligned} \tag{2.12}$$

Contracting the first two identities with $\mathbf{f}(x)^\top$ and $\mathbf{p}(qx)$, and using (2.8) and (2.10), gives

$$h(qx) = \mathbf{f}(x)^\top K_1(x)\mathbf{p}(x), \quad qxh(q^2x) = \mathbf{f}(x)^\top K_2(x)\mathbf{p}(x).$$

The same contraction of the last identity gives qx times the left side of (2.5) with $g = h$. Cancelling qx shows that h satisfies that equation.

The defining sums give

$$a(0) = R_0, \quad [x]a = \frac{qR_1}{1-q}, \quad F(x, x^3) = 1 + \frac{q}{1-q}x + O(x^2).$$

The other summand in h starts at x^2 . Hence h has the two initial coefficients in (2.6). The initial two coefficients and the recurrence agree, so $h = g$.

The matrix K_1 and (2.11) now give (2.2). For the last identity, termwise subtraction gives $g(x) - g(qx) = qxG(q^2x, q^{-3})$. The relation $R_{-j-1} = R_{-j} + q^{-3j}R_{1-j}$, from (2.4), then gives

$$qxG(q^{-1}x, q^3) = g(x) - g(qx) - qxg(q^2x).$$

Finally, direct multiplication gives

$$K_0 - K_1 - K_2 = qx \begin{pmatrix} 0 & 0 & 0 \\ 0 & x & q^2x^2 \\ 1 & qx & 0 \end{pmatrix}.$$

Contracting with $\mathbf{f}(x)^\top$ and $\mathbf{p}(x)$, using (2.11), and cancelling qx proves (2.3). $\square$

At $x = 1$, Proposition 2.1 yields

$$G(1, 1) = A^2 + qBC, \quad G(q^{-1}, q^3) = B^2 + AC, \quad G(q, 1) = AB - qC^2. \tag{2.13}$$

The identities (2.12) define a bilinear intertwiner from the two difference systems for F to the scalar difference equation for G . The quadratic expressions in (2.13) arise by specializing this intertwiner at $x = 1$.

3 Casoratian

Define the q -Airy function with base p by

$$\text{Ai}_p(X) = \sum_{n \geq 0} \frac{p^{n(n-1)/2} X^n}{(p^2; p^2)_n}.$$

Its elementary difference equation is

$$\text{Ai}_p(X) - \text{Ai}_p(p^2 X) = X \text{Ai}_p(pX). \quad (3.1)$$

Consequently the function

$$\omega(X) = \text{Ai}_p(X) \text{Ai}_p(-pX) + \text{Ai}_p(-X) \text{Ai}_p(pX)$$

satisfies $\omega(pX) = \omega(X)$. Iterating towards zero gives

$$\omega(X) = \omega(0) = 2. \quad (3.2)$$

Let t satisfy $t^4 = q$, and define

$$\mathcal{U}(t; x) = \sum_{m, n \geq 0} \frac{(-1)^m t^{m^2 + 6mn + 3n^2 + 6n} x^m}{(t^4; t^4)_m (t^{12}; t^{12})_n},$$

$$\mathcal{A}(x) = \frac{\mathcal{U}(t; x) + \mathcal{U}(-t; x)}{2}, \quad \mathcal{B}(x) = \frac{\mathcal{U}(t; x) - \mathcal{U}(-t; x)}{2t}, \quad \mathbf{r}(x) = (\mathcal{A}(x), \mathcal{B}(x)).$$

Every exponent of $\mathcal{U}$ is either 0 or 1 (mod 4). Thus $\mathcal{A}, \mathcal{B}$ are independent of the choice of fourth root after the parity separation. Define the two rows

$$(A_1, B_1) = \mathbf{r}(1), \quad (A_2, B_2) = \mathbf{r}(q). \quad (3.3)$$

Let $J_m = \text{Ai}_{t^6}(t^{6m+9})$. Then $[x^m] \mathcal{U}(t; x) = (-1)^m t^{m^2} J_m / (q; q)_m$. Equation (3.1) gives $J_{m-2} - J_m = t^{6m-3} J_{m-1}$, so these coefficients satisfy the same recurrence as g_m in Section 2. Both $\mathcal{A}$ and $\mathcal{B}$ therefore satisfy (2.5). Moreover, (3.2) at $p = t^6, X = t^9$ gives

$$\det \begin{pmatrix} \mathbf{r}(0) \\ \mathbf{r}'(0) \end{pmatrix} = -\frac{\text{Ai}_{t^6}(t^9) \text{Ai}_{t^6}(-t^{15}) + \text{Ai}_{t^6}(-t^9) \text{Ai}_{t^6}(t^{15})}{2(1-q)} = -\frac{1}{1-q}. \quad (3.4)$$

For each positive integer j , define

$$W_j(x) = \det \begin{pmatrix} \mathbf{r}(x) \\ \mathbf{r}(q^j x) \end{pmatrix}.$$

Equation (3.4) gives $W_1(x) = x + O(x^2)$. Put $H(x) = W_1(x)/x$, with its removable value $H(0) = 1$. Taking two-row determinants in (2.5) gives

$$q^2 x W_2(qx) = q W_1(x) - (1 - q^2 x) W_1(qx),$$

$$q W_2(x) = (1 + q + q^2 x^2) W_1(qx) + q^2 x W_1(q^2 x).$$

Eliminating W_2 , substituting $W_1(x) = xH(x)$, and cancelling qx gives

$$H(x) = (1 - q^2 x) H(qx) + q^2 x (1 + q + q^4 x^2) H(q^2 x) + q^6 x^2 H(q^3 x). \quad (3.5)$$

For each positive degree j , the new coefficient is multiplied by $1 - q^j \neq 0$. The analytic solution with $H(0) = 1$ is unique. Equation (2.9), with x replaced by qx , shows that $F(q^2 x, q^3 x^3)$ is this solution. Therefore

$$\det \begin{pmatrix} \mathcal{A}(x) & \mathcal{B}(x) \\ \mathcal{A}(qx) & \mathcal{B}(qx) \end{pmatrix} = x F(q^2 x, q^3 x^3). \quad (3.6)$$

In particular, at $x = 1$,

$$\det \begin{pmatrix} A_1 & B_1 \\ A_2 & B_2 \end{pmatrix} = F(q^2, q^3) = C. \quad (3.7)$$

4 Connection formula

Define the theta function by

$$\vartheta_p^+(z) = \sum_{k \in \mathbb{Z}} p^{k(k-1)/2} z^k.$$

Define the Ramanujan function by

$$\mathcal{R}_q(Z) = \sum_{n \geq 0} \frac{q^{n^2} (-Z)^n}{(q; q)_n}.$$

The connection formula [12, Section 1, Theorem], with the q -Airy convention of Section 3, is

$$\vartheta_p^+(X/p) \text{Ai}_p(-X) + \vartheta_p^+(-X/p) \text{Ai}_p(X) = 2(p^2; p^2)_\infty \mathcal{R}_{p^2}(-p^3/X^2). \quad (4.1)$$

It holds for $X \neq 0$. The factor in the cited formula is $(p, -1; p)_\infty = 2(p^2; p^2)_\infty$.

The definitions of the parity rows give

$$\mathcal{U}(\pm t; 1) = A_1 \pm tB_1, \quad \mathcal{U}(\pm t; q) = A_2 \pm tB_2. \quad (4.2)$$

To evaluate $\mathcal{U}(t; q^j)$ for $j = 0, 1$, first use $p = t^2$, $X = t^{6n+4j+1}$. Quasiperiodicity gives

$$\vartheta_{t^2}^+(\pm t^{6n+4j-1}) = (\pm 1)^n L_n \vartheta_{t^2}^+(\pm t^{4j-1}), \quad L_n = t^{-9n^2+(6-12j)n}.$$

Multiply (4.1) by $t^{3n^2+6n}/((t^{12}; t^{12})_n L_n)$, and sum over $n \geq 0$. The exponent on the right is

$$4m^2 - 12mn + 12n^2 + (4 - 8j)m + 12jn.$$

At $j = 0, 1$, respectively, it gives $\mathcal{G}(q, 1)$ and $\mathcal{G}(q^{-1}, q^3)$. The symmetry $\vartheta_{t^2}^+(\pm t^3) = \vartheta_{t^2}^+(\pm t^{-1})$ therefore gives

$$\begin{aligned} \vartheta_{t^2}^+(t^{-1})\mathcal{U}(t; 1) + \vartheta_{t^2}^+(-t^{-1})\mathcal{U}(-t; 1) &= 2(q; q)_\infty \mathcal{G}(q, 1), \\ \vartheta_{t^2}^+(t^{-1})\mathcal{U}(t; q) + \vartheta_{t^2}^+(-t^{-1})\mathcal{U}(-t; q) &= 2(q; q)_\infty \mathcal{G}(q^{-1}, q^3). \end{aligned} \quad (4.3)$$

Next use $p = t^6$, $X = t^{6m+9}$. In this case

$$\vartheta_{t^6}^+(\pm t^{6m+3}) = (\pm 1)^m t^{-3m^2} \vartheta_{t^6}^+(\pm t^3).$$

Multiply by $t^{4m^2+4jm}/(t^4; t^4)_m$, and sum over $m \geq 0$. The exponent on the right is $4m^2 - 12mn + 12n^2 + 4jm$. Thus

$$\begin{aligned} \vartheta_{t^6}^+(-t^3)\mathcal{U}(t; 1) + \vartheta_{t^6}^+(t^3)\mathcal{U}(-t; 1) &= 2(q^3; q^3)_\infty \mathcal{G}(1, 1), \\ \vartheta_{t^6}^+(-t^3)\mathcal{U}(t; q) + \vartheta_{t^6}^+(t^3)\mathcal{U}(-t; q) &= 2(q^3; q^3)_\infty \mathcal{G}(q, 1). \end{aligned} \quad (4.4)$$

Define

$$\theta(q) = \sum_{n \in \mathbb{Z}} q^{n^2}, \quad \varphi(q) = \sum_{n \in \mathbb{Z}} q^{n(n-1)}.$$

Even and odd index separation gives

$$\vartheta_{t^2}^+(\pm t^{-1}) = \varphi(q) \pm t^{-1}\theta(q), \quad \vartheta_{t^6}^+(\pm t^3) = \theta(q^3) \pm t^3\varphi(q^3).$$

Using (4.2) in (4.3)–(4.4) consequently proves

$$\begin{pmatrix} A_1 & B_1 \\ A_2 & B_2 \end{pmatrix} \begin{pmatrix} \varphi(q) & \theta(q^3) \\ \theta(q) & -q\varphi(q^3) \end{pmatrix} = \begin{pmatrix} (q; q)_\infty \mathcal{G}(q, 1) & (q^3; q^3)_\infty \mathcal{G}(1, 1) \\ (q; q)_\infty \mathcal{G}(q^{-1}, q^3) & (q^3; q^3)_\infty \mathcal{G}(q, 1) \end{pmatrix}. \quad (4.5)$$

5 A cubic identity for the products

Define

$$\mathfrak{a}(q) = \sum_{r,s \in \mathbb{Z}} q^{r^2+rs+s^2}, \quad L(q) = \sum_{n \geq 1} \frac{q^n}{(1-q^n)^2}.$$

We use the square identity [10, Theorem 7.2]

$$\mathfrak{a}(q)^2 = 1 + 12L(q) - 36L(q^3) \quad (5.1)$$

and the trisection identity [1, equation (2.4), with q replaced by q^3]

$$(\mathfrak{a}(q) - \mathfrak{a}(q^3))(q^3; q^3)_\infty = 6q(q^9; q^9)_\infty^3. \quad (5.2)$$

Write the Jacobi product as

$$j(z; q) = (q, z, q/z; q)_\infty. \quad (5.3)$$

It satisfies

$$j(qz; q) = -z^{-1}j(z; q), \quad j(z^{-1}; q) = -z^{-1}j(z; q), \quad j(q/z; q) = j(z; q). \quad (5.4)$$

The Weierstrass addition formula gives

$$\begin{aligned} j(xz; \rho)j(x/z; \rho)j(y; \rho)^2 - j(yz; \rho)j(y/z; \rho)j(x; \rho)^2 \\ = \frac{y}{z}j(xy; \rho)j(x/y; \rho)j(z; \rho)^2. \end{aligned} \quad (5.5)$$

For an auxiliary variable u and nome ρ , define

$$\Sigma(u; \rho) = \sum_{n \geq 0} \left\{ \frac{u\rho^n}{(1-u\rho^n)^2} + \frac{\rho^{n+1}/u}{(1-\rho^{n+1}/u)^2} \right\}.$$

Expanding the paired factors of the product at $z = 1 + \varepsilon$ gives

$$j(u(1+\varepsilon); \rho)j(u/(1+\varepsilon); \rho) = j(u; \rho)^2(1 - \Sigma(u; \rho)\varepsilon^2) \pmod{\varepsilon^3}, \quad (5.6)$$

since $(1-bz)(1-b/z)/(1-b)^2 = 1 - b(z+z^{-1}-2)/(1-b)^2$ and $z+z^{-1}-2 = \varepsilon^2 \pmod{\varepsilon^3}$.

Proposition 5.1. *The three products satisfy*

$$\mathcal{N}(\text{KR}_1, \text{KR}_2, \text{KR}_3) = \frac{\mathfrak{a}(q)}{(q; q)_\infty (q^3; q^3)_\infty}. \quad (5.7)$$

Proof. For $1 \leq r < 9$, write $J_r = j(q^r; q^9)$ and $\Lambda_r = \Sigma(q^r; q^9)$. Splitting the products by residue classes gives

$$\begin{aligned} J_3 &= (q^3; q^3)_\infty, & J_1 J_2 J_4 &= \frac{(q; q)_\infty (q^9; q^9)_\infty^3}{(q^3; q^3)_\infty}, \\ (\text{KR}_1, \text{KR}_2, \text{KR}_3) &= \frac{(q^9; q^9)_\infty^2}{(q^3; q^3)_\infty} \left(\frac{1}{J_1}, \frac{1}{J_2}, \frac{1}{J_4} \right), \\ \text{KR}_1 \text{KR}_2 \text{KR}_3 &= \frac{(q^9; q^9)_\infty^3}{(q; q)_\infty (q^3; q^3)_\infty^2}. \end{aligned}$$

In (5.5), substitute $(\rho, x, y) = (q^9, q^r, q^3)$ and $z = 1 + \varepsilon$. The second coefficient, using (5.6), gives

$$J_r^2 (q^3; q^3)_\infty^2 (\Lambda_r - \Lambda_3) = -q^3 (q^9; q^9)_\infty^6 j(q^{r+3}; q^9) j(q^{r-3}; q^9).$$

For $r = 1, 2, 4$, reflection and the preceding product relations simplify these identities to

$$\begin{aligned}\Lambda_1 - \Lambda_3 &= q(q; q)_\infty (q^9; q^9)_\infty^3 \text{KR}_1^3, \\ \Lambda_2 - \Lambda_3 &= q^2(q; q)_\infty (q^9; q^9)_\infty^3 \text{KR}_2^3, \\ \Lambda_4 - \Lambda_3 &= -q^3(q; q)_\infty (q^9; q^9)_\infty^3 \text{KR}_3^3.\end{aligned}$$

Their sum has left side $L(q) - 4L(q^3) + 3L(q^9)$. Multiplying by $(q^3; q^3)_\infty^2$ and using the product of the three KR_i gives

$$\begin{aligned}(q^3; q^3)_\infty^2 (L(q) - 4L(q^3) + 3L(q^9)) \\ = q(q; q)_\infty (q^3; q^3)_\infty^2 (q^9; q^9)_\infty^3 \mathcal{N}(\text{KR}_1, \text{KR}_2, \text{KR}_3) - 3q^2(q^9; q^9)_\infty^6.\end{aligned}\tag{5.8}$$

It remains to evaluate the Lambert expression. Subtracting (5.1) at q^3 from the same identity at q yields

$$12(L(q) - 4L(q^3) + 3L(q^9)) = (\mathfrak{a}(q) - \mathfrak{a}(q^3))(\mathfrak{a}(q) + \mathfrak{a}(q^3)).$$

Applying (5.2) and writing the sum of the two theta series as $2\mathfrak{a}(q) - (\mathfrak{a}(q) - \mathfrak{a}(q^3))$ gives

$$(q^3; q^3)_\infty^2 (L(q) - 4L(q^3) + 3L(q^9)) = q\mathfrak{a}(q)(q^3; q^3)_\infty (q^9; q^9)_\infty^3 - 3q^2(q^9; q^9)_\infty^6.$$

Comparison with (5.8) and cancellation proves the proposition. $\square$

6 Proof of the main theorem

Taking determinants in (4.5) and applying Proposition 5.1 gives the following equalities.

Lemma 6.1 (Equality of the cubic norms). *As identities of formal power series,*

$$\mathcal{N}(\mathbf{A}, \mathbf{B}, \mathbf{C}) = \frac{\mathfrak{a}(q)}{(q; q)_\infty (q^3; q^3)_\infty} = \mathcal{N}(\text{KR}_1, \text{KR}_2, \text{KR}_3).\tag{6.1}$$

Proof. Splitting the summation lattice according to the parity of s gives

$$\mathfrak{a}(q) = \theta(q)\theta(q^3) + q\varphi(q)\varphi(q^3).\tag{6.2}$$

Indeed, the substitutions $(r, s) = (\ell - k, 2k)$ and $(r, s) = (\ell - k, 2k - 1)$ give the exponents $\ell^2 + 3k^2$ and $\ell(\ell - 1) + 3k(k - 1) + 1$, respectively. Taking determinants directly in (4.5), and using (3.7) and (6.2), gives

$$\mathbf{G}(q, 1)^2 - \mathbf{G}(1, 1)\mathbf{G}(q^{-1}, q^3) = -\frac{\mathbf{C}\mathfrak{a}(q)}{(q; q)_\infty (q^3; q^3)_\infty}.$$

Substituting (2.13) yields

$$(\mathbf{A}\mathbf{B} - q\mathbf{C}^2)^2 - (\mathbf{A}^2 + q\mathbf{B}\mathbf{C})(\mathbf{B}^2 + \mathbf{A}\mathbf{C}) = -\frac{\mathbf{C}\mathfrak{a}(q)}{(q; q)_\infty (q^3; q^3)_\infty}.$$

The polynomial identity

$$(XY - qZ^2)^2 - (X^2 + qYZ)(Y^2 + XZ) = -Z\mathcal{N}(X, Y, Z)\tag{6.3}$$

therefore proves the first equality in (6.1). The cancellation of $\mathbf{C}$ is valid because its constant coefficient is 1.

The second equality is Proposition 5.1. $\square$

One further input supplies the direction of every possible discrepancy between the two triples.

Lemma 6.2 (Coefficientwise lower bounds). *The three differences satisfy*

$$A - KR_1, \quad B - KR_2, \quad C - KR_3 \in q\mathbb{Z}_{\geq 0}[[q]]. \quad (6.4)$$

Proof. Let $\mathcal{D}_i$, $i = 1, 2, 3$, be the partition sets in the introduction. Kurşungöz proves that the generating functions $\sum_{\lambda \in \mathcal{D}_i} q^{|\lambda|}$ are A, B, C , respectively [9, Theorems 7–9, equations (1), (38), (85)]. In his two-variable formulas, the length variable is set to 1. The three exponents, in the present indices (m, n) , are

$$m^2 + 3mn + 3n^2, \quad m^2 + 3mn + 3n^2 + m + 3n, \quad m^2 + 3mn + 3n^2 + 2m + 3n.$$

These are exactly the exponents in the definitions of A, B, C .

Tsuchioka's spanning theorem implies that, for every $N \geq 0$, the number of partitions of N in $\mathcal{D}_i$ is at least the number of partitions of N into parts congruent to $2^{i-1}, 3, 6, 9 - 2^{i-1} \pmod{9}$ [13, Corollary 1.4]. The latter generating functions are KR_1, KR_2, KR_3 . All six series have constant coefficient 1, proving (6.4). $\square$

Lemma 6.3 (Rigidity under coefficientwise lower bounds). *Let $X, Y, Z, x, y, z \in 1 + q\mathbb{Z}[[q]]$. If $X - x, Y - y, Z - z$ have nonnegative coefficients and $\mathcal{N}(X, Y, Z) = \mathcal{N}(x, y, z)$, then $(X, Y, Z) = (x, y, z)$.*

Proof. Write $d_1 = X - x$, $d_2 = Y - y$, and $d_3 = Z - z$. Telescoping the cubic and mixed terms gives the exact identity

$$\begin{aligned} \mathcal{N}(X, Y, Z) - \mathcal{N}(x, y, z) &= d_1(X^2 + Xx + x^2 + 3qYZ) \\ &\quad + qd_2(Y^2 + Yy + y^2 + 3xZ) \\ &\quad + qd_3(3xy - q(Z^2 + Zz + z^2)). \end{aligned} \quad (6.5)$$

The three expressions in parentheses have constant coefficients 3, 6, 3, respectively. Suppose that at least one d_i is nonzero, and define

$$\nu = \min\{\text{ord}_q d_1, 1 + \text{ord}_q d_2, 1 + \text{ord}_q d_3\}, \quad \text{ord}_q 0 = +\infty.$$

Each summand of order ν on the right of (6.5) has a positive coefficient at that order. Every other summand has greater order. Thus the coefficient of q^ν is positive, contradicting the equality of the norms. All three differences vanish. $\square$

Proof of Theorem 1.1. Apply Lemma 6.3 to the equal norms in Lemma 6.1 and the lower bounds in Lemma 6.2. $\square$

Use of AI

I would first like to describe how I came to ask AI to solve this conjecture. I learned about the conjecture while I was a student in the Department of Mathematical and Computing Science at Tokyo Institute of Technology (now Institute of Science Tokyo), and I remember hearing about it from Shunsuke Tsuchioka. I do not remember exactly when I first heard of it, but hearing about it in Tsuchioka's lectures (on Rogers–Ramanujan identities and related topics), which I attended via Zoom in 2020, left a lasting impression on me.

When I saw the conjecture, I noticed that the q -series on the sum side could be realized as partition q -series of Kato and Terashima [7] in the context of cluster algebras, which I was

studying at the time (Terashima was then my supervisor). This realization could be used to derive related dilogarithm identities, but it did not provide a way to prove the original q -series identities. Ever since then, the conjecture had remained on my mind. Its connection with cluster algebras was also one of the main motivations for my later paper [11]. In that paper, I stated the modular transformation formulas for the products without proof. I had in fact obtained a proof of the transformation formulas by expressing the products as double theta series, but had not written it up because it lay outside the main subject of the paper and I thought it might be useful for the Kanade–Russell conjecture itself. Wang and Zhang [15] later proved these transformation formulas with a shorter proof than mine, so, in retrospect, I think this worked out well.

Because this history had kept the problem on my mind, I tried asking each new AI model whether it could solve the conjecture. At first, I gave the models my own previous attempts (mostly recorded in SageMath files) and asked whether they could solve it. Eventually, the files generated during their attempts came to outnumber my own. The first model to make meaningful progress was GPT-5.6 Sol, which discovered on July 22, 2026 that the series on the sum side could be expressed as a Casoratian, as in (3.6). This did not yet lead to a complete proof. When GPT-6 Astra became available in September 2026, I asked, as usual, whether it could solve the conjecture. This time, the attempt led to a complete proof. The key discovery was the equality of cubic norms in Lemma 6.1. This cubic already arises naturally from the modular transformation formulas for the products that I had stated in my earlier paper, but I had not thought to consider it.

The reason this cubic arises naturally can be explained as follows. Put $q = e^{2\pi i\tau}$ with $\text{Im } \tau > 0$, $\zeta = e^{2\pi i/9}$, and $\omega = \zeta^3$. The column vector

$$f(\tau) = (q^{-1/18}\text{KR}_1, q^{5/18}\text{KR}_2, q^{11/18}\text{KR}_3)^T$$

satisfies $f(-1/(3\tau)) = Sf(\tau)$ [11, 15], where

$$S = \begin{pmatrix} \alpha_1 & \alpha_2 & \alpha_4 \\ \alpha_2 & -\alpha_4 & -\alpha_1 \\ \alpha_4 & -\alpha_1 & \alpha_2 \end{pmatrix}, \quad \alpha_j = \frac{1}{2\sqrt{3}\sin(j\pi/9)}.$$

For the linear forms $L_j : \mathbb{C}^3 \rightarrow \mathbb{C}$, $L_j(X, Y, Z) = X + \omega^j Y - \omega^{2j} Z$ ($j = 0, 1, 2$), we have

$$\begin{aligned} L_0 L_1 L_2 &= X^3 + Y^3 - Z^3 + 3XYZ, \\ L_0 \circ S &= L_0, \quad L_1 \circ S = \zeta L_2, \quad L_2 \circ S = \zeta^{-1} L_1. \end{aligned}$$

Thus their product is invariant under S . Evaluating it at $f(\tau)$ gives

$$L_0(f(\tau))L_1(f(\tau))L_2(f(\tau)) = q^{-1/6}\mathcal{N}(\text{KR}_1, \text{KR}_2, \text{KR}_3).$$

This recovers the cubic $\mathcal{N}$.

GPT-6 Astra found that the same value could also be obtained from the sums, establishing a clear relation between the sum and product sides. It then combined this equality with Tsuchioka’s coefficientwise inequalities (Lemma 6.2) as the final piece of the proof, thereby proving that the two sides coincide.

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